The Dataverse Network

An open-source Application for Sharing, Discovering and Preserving Data

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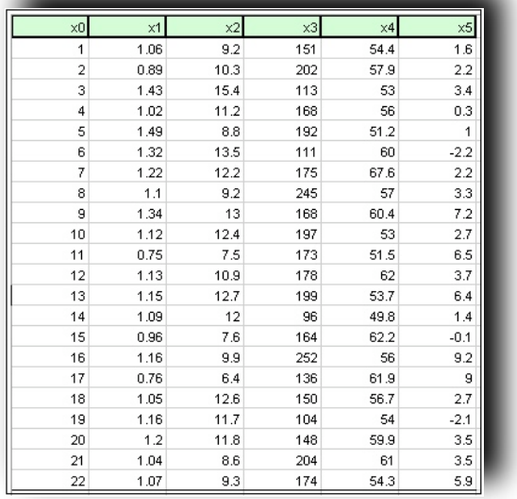
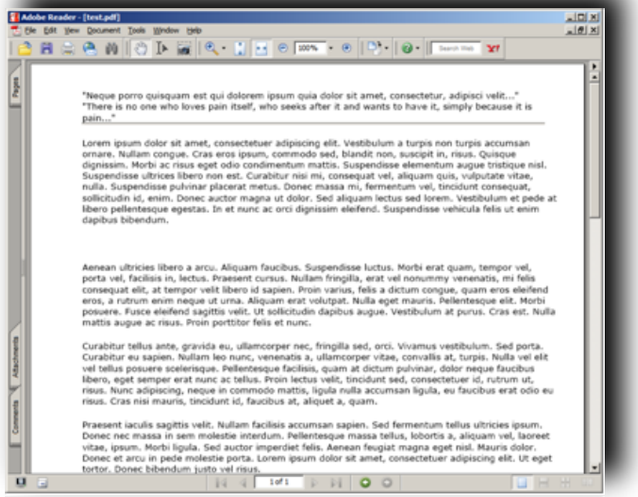
Problems with traditional ways to store research data



x0	x1	x2	x3	x4	x5
1	1.06	9.2	151	54.4	1.6
2	0.89	10.3	202	57.9	2.2
3	1.43	15.4	113	53	3.4
4	1.02	11.2	168	56	0.3
5	1.49	8.8	192	51.2	1
6	1.32	13.5	111	60	-2.2
7	1.22	12.2	175	67.6	2.2
8	1.1	9.2	245	57	3.3
9	1.34	13	168	60.4	7.2
10	1.12	12.4	197	53	2.7
11	0.75	7.5	173	51.5	6.5
12	1.13	10.9	178	62	3.7
13	1.15	12.7	199	53.7	6.4
14	1.09	12	96	49.8	1.4
15	0.96	7.6	164	62.2	-0.1
16	1.16	9.9	252	56	9.2
17	0.76	6.4	136	61.9	9
18	1.05	12.6	150	56.7	2.7
19	1.16	11.7	104	54	-2.1
20	1.2	11.8	148	59.9	3.5
21	1.04	8.6	204	61	3.5
22	1.07	9.3	174	54.3	5.9

Data sets



Text

Images

Keep it in my laptop → Hard to share, not safe

Send it to an archive → Author doesn't get enough credit

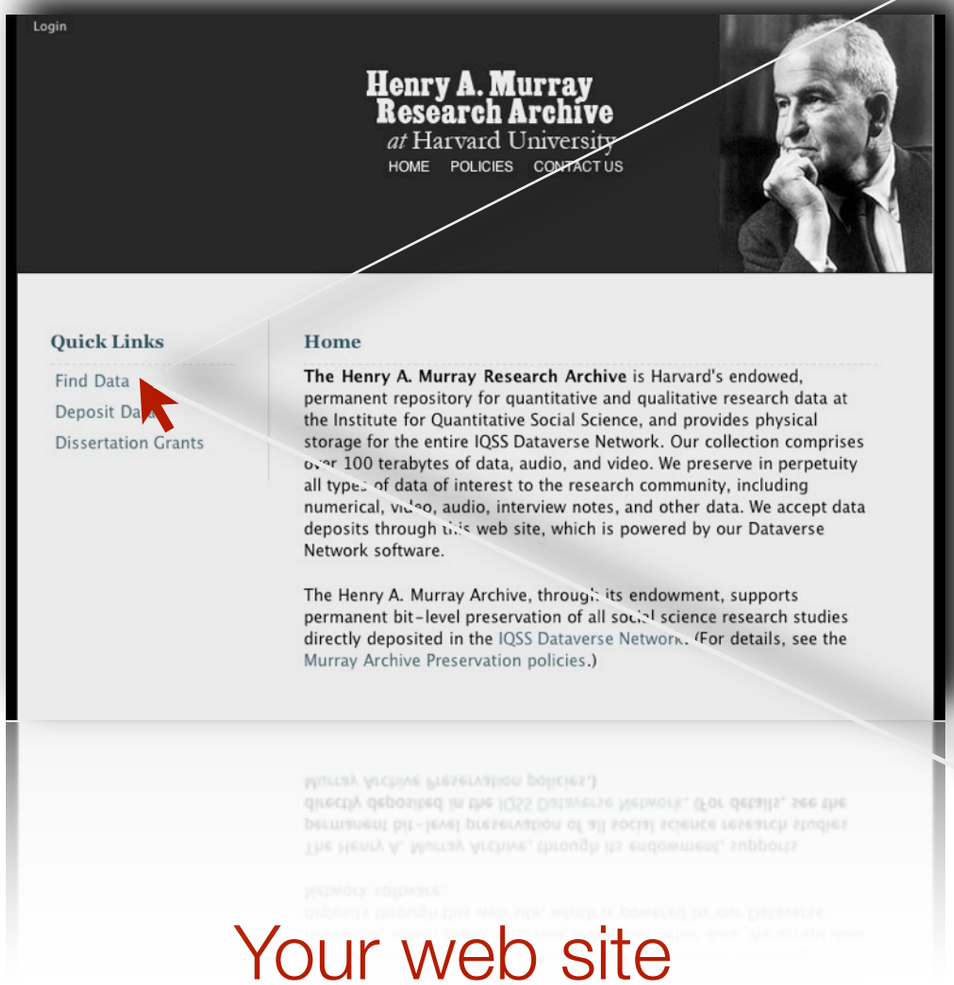
Post it in my web site → No professional archiving services (backups, recovering, cataloging, preservation, etc)

Solving the problems through technology

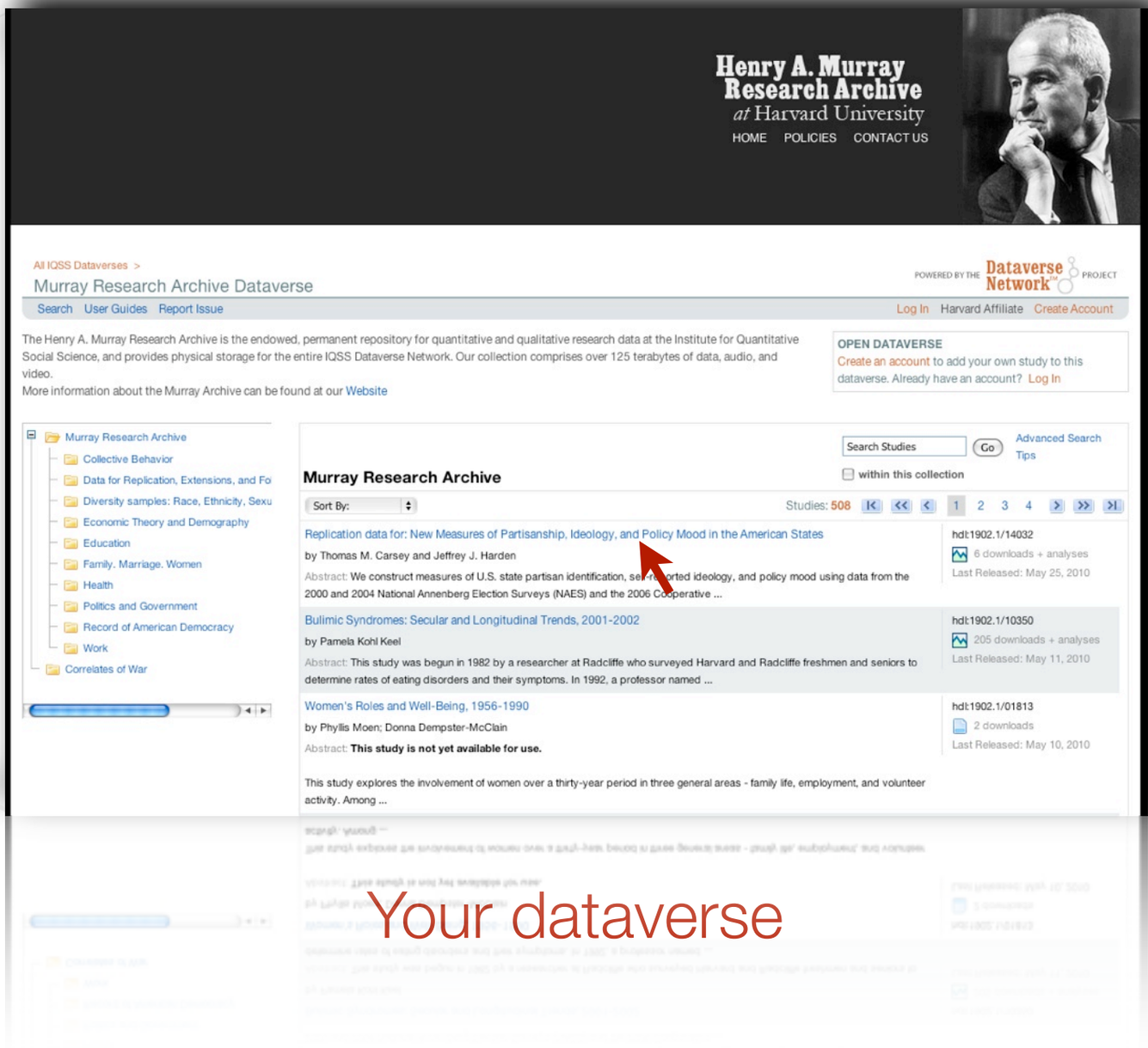
We've developed a system that:

- ☑ Gives ownership and recognition to author, distributor and publisher
- ☑ Generates a persistent data citation
- ☑ Converts data sets to a preservable and verifiable format
- ☑ Distributes data to the public, but also supports restricted access
- ☑ Indexes all metadata for quick data discovery
- ☑ Supports subsetting and analysis for (some) data files
- ☑ Makes it easy to use and maintain
- ☑ Provides you legal protection
- ☑ Inter-operates with other systems using standards

Gives ownership and recognition



Your web site



Your dataverse

☑ Gives ownership and recognition

All IQSS Dataverses >

Murray Research Archive Dataverse

POWERED BY THE Dataverse Network™ PROJECT

Search User Guides Report Issue Log In Harvard Affiliate Create Account

REPLICATION DATA FOR: NEW MEASURES OF PARTISANSHIP, IDEOLOGY, AND POLICY MOOD IN THE AMERICAN STATES

< View Previous Study Listing

hdl:1902.1/14032 UNF:5:EKgHvTNfkkS86dNzABIhNw==

Version: 2 – Released: Tue May 25 10:53:02 EDT 2010

Cataloging Information | Documentation, Data and Analysis | User Comments | Versions

How to Cite Why cite?

In order to credit those who developed and distributed the data, select either "print" or "web" citation format, copy the citation generated and paste into your document or website. This citation contains a persistent identifier (and in some cases, a Numerical Fingerprint) for permanent reference. Include the citation wherever the data is referenced.

Please use this citation

```
Thomas M. Carsey and Jeffrey J. Harden, 2010, "Replication data for: New Measures of
Partisanship, Ideology, and Policy Mood in the American States",
http://hdl.handle.net/1902.1/14032 UNF:5:EKgHvTNfkkS86dNzABIhNw== Murray Research
Archive [Distributor] V2 [Version]
```

Credit to the author

Credit to the distributor

Citation Format Print

Citation Information

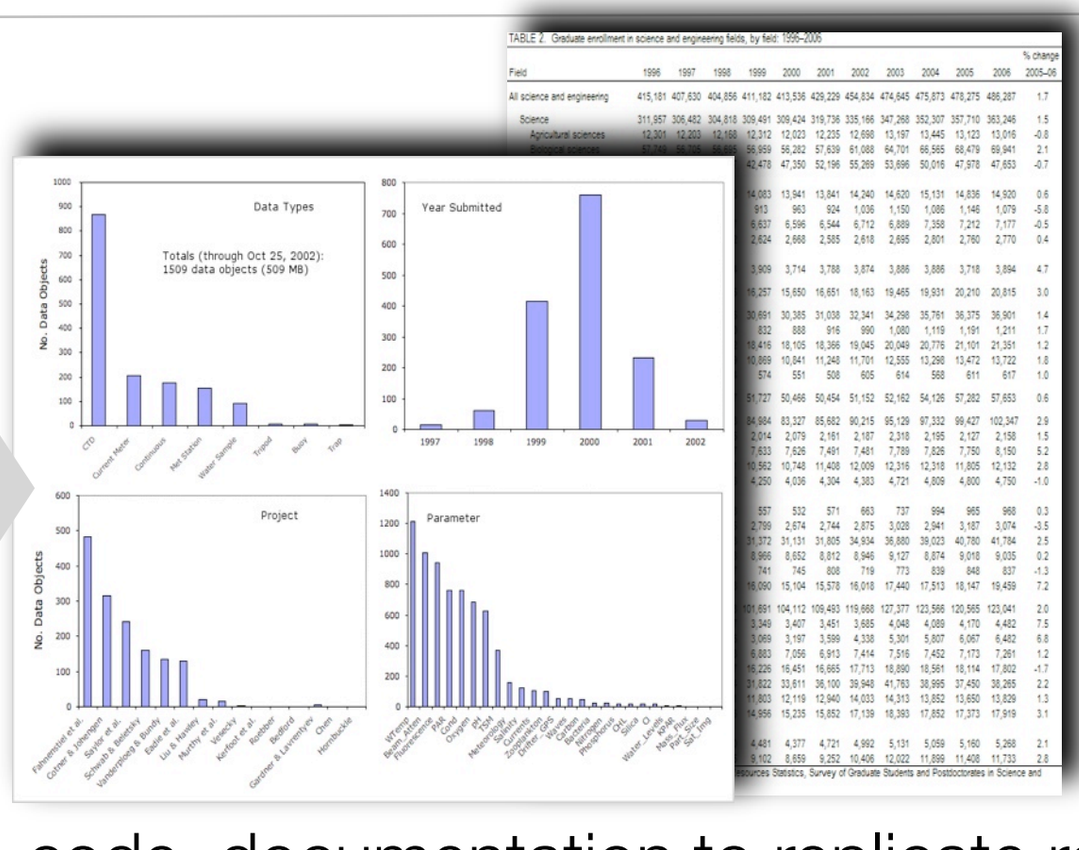
Study Global Id	hdl:1902.1/14032
Authors	Thomas M. Carsey and Jeffrey J. Harden (University of North Carolina at Chapel Hill)
Production Date	2010
Distributor	Murray Research Archive (Henry A. Murray Research Archive at Harvard University)
Distributor Contact	Jeffrey J. Harden (University of North Carolina at Chapel Hill), jjharden@unc.edu
Deposit Date	January 18, 2010
Replication For	Carsey, Thomas M. and Jeffrey J. Harden. 2010. "New Measures of Partisanship, Ideology, and Policy Mood in the American States." State Politics and Policy Quarterly 10(2):136-156.
Provenance	Murray Research Archive Dataverse

Abstract and Scope

Abstract	We construct measures of U.S. state partisan identification, self-reported ideology, and policy mood using data from the 2000 and 2004 National Annenberg Election Surveys (NAES) and the 2006 Cooperative Congressional Election Study (CCES). These measures improve on existing methods for estimating state-level preferences because the surveys provide larger state samples without pooling across years. After detailing our methods for constructing the measures, we assess their validity through comparisons with measures already in use by scholars

Persistent Citation for Data

Formal data citation



Journal article, book

Dataset, code, documentation to replicate results

A persistent identifier and url which never changes (hdl:1902.2/7768):
http://handle.net/1902.2/7768

Universal Numerical Fingerprint to verify data set:
UNF:3:upor0mnBeXa9fujUkPqNNA==

Convert Data to a Preservable and Verifiable Format

Data Set File

Metadata and Format Information

$$\begin{pmatrix} 1 & 4 & 4 & 21 & \cdots & 121 \\ 1 & 2 & 2 & 91 & \cdots & 212 \\ 1 & 9 & 2 & 72 & \cdots & 104 \\ 0 & 2 & 2 & 2 & \cdots & 321 \\ 1 & 6 & 2 & 12 & \cdots & 204 \\ 1 & 9 & 4 & 52 & \cdots & 311 \\ 0 & 3 & 2 & 23 & \cdots & 92 \\ 0 & 2 & 5 & 91 & \cdots & 212 \\ 0 & 5 & 8 & 91 & \cdots & 91 \\ 1 & 9 & 1 & 72 & \cdots & 104 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 2 & 91 & \cdots & 212 \end{pmatrix}$$

1. Extract variable name, description and summary statistics
2. Convert data set to a preservation format, independent of the software package
3. Apply a cryptographic algorithm to canonical format
4. Get alphanumeric string based on semantic contents of the digital object:
   - uniquely summarizes the contents,
   - but does not convey its information

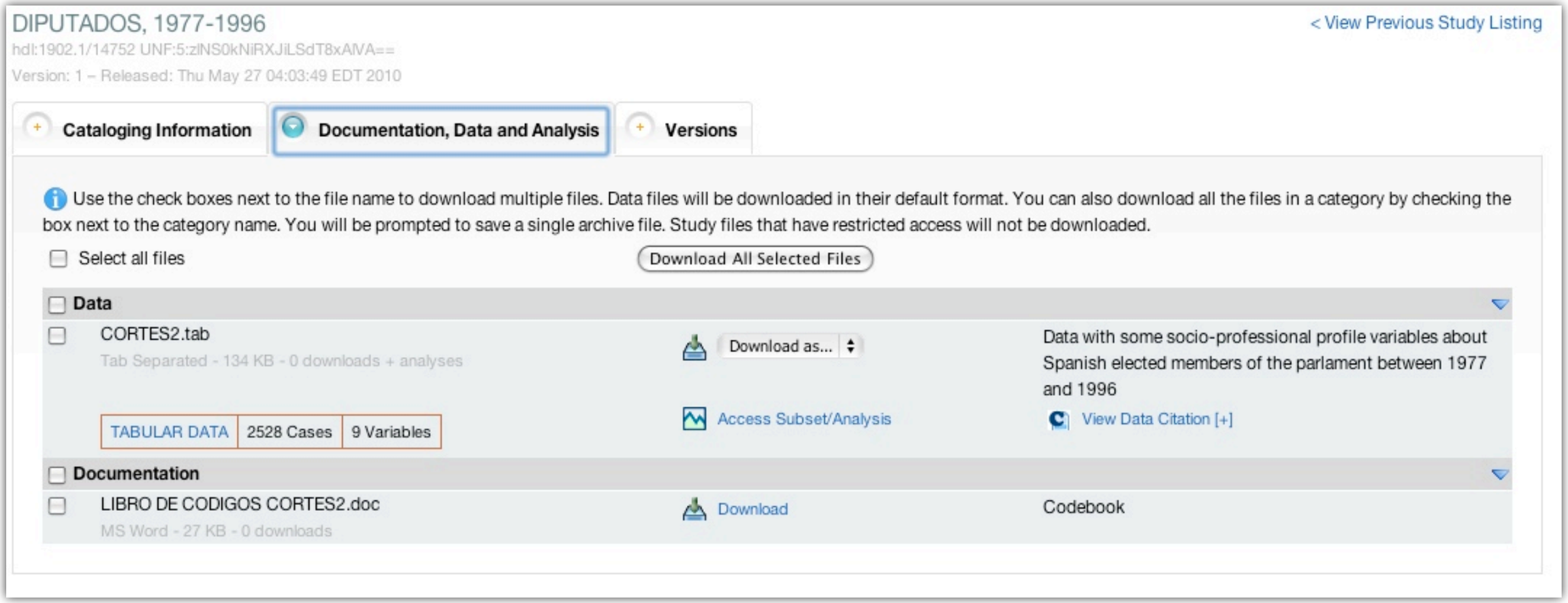
UNF:5:EKgHvTNfkkS86dNzABIhNw==

Change content → Changes UNF

Change format → Doesn't change UNF

☑ Distributes to the public, but also supports restricted access

Studies and files can be completely public ...



Everybody can explore the cataloging information (metadata), download data and documentation files, and subset and analyze data if available.

☑ Distributes to the public, but also supports restricted access

... or some files can be restricted ...

THE ASPIRATIONS AND EXPERIENCES OF UNDERGRADUATES, 2000

< View Previous Study Listing

hdl:1902.1/00092 UNF:3:0sjRWertkBtT9wlgcuP4Fg==

Version: 1 – Released: Mon May 10 13:02:44 EDT 2010

Cataloging Information | Documentation, Data and Analysis | User Comments | Versions

Use the check boxes next to the file name to download multiple files. Data files will be downloaded in their default format. You can also download all the files in a category by checking the box next to the category name. You will be prompted to save a single archive file. Study files that have restricted access will not be downloaded.

Access to some files is restricted, and those files are not available for downloading. Check the Terms of Use for more information.

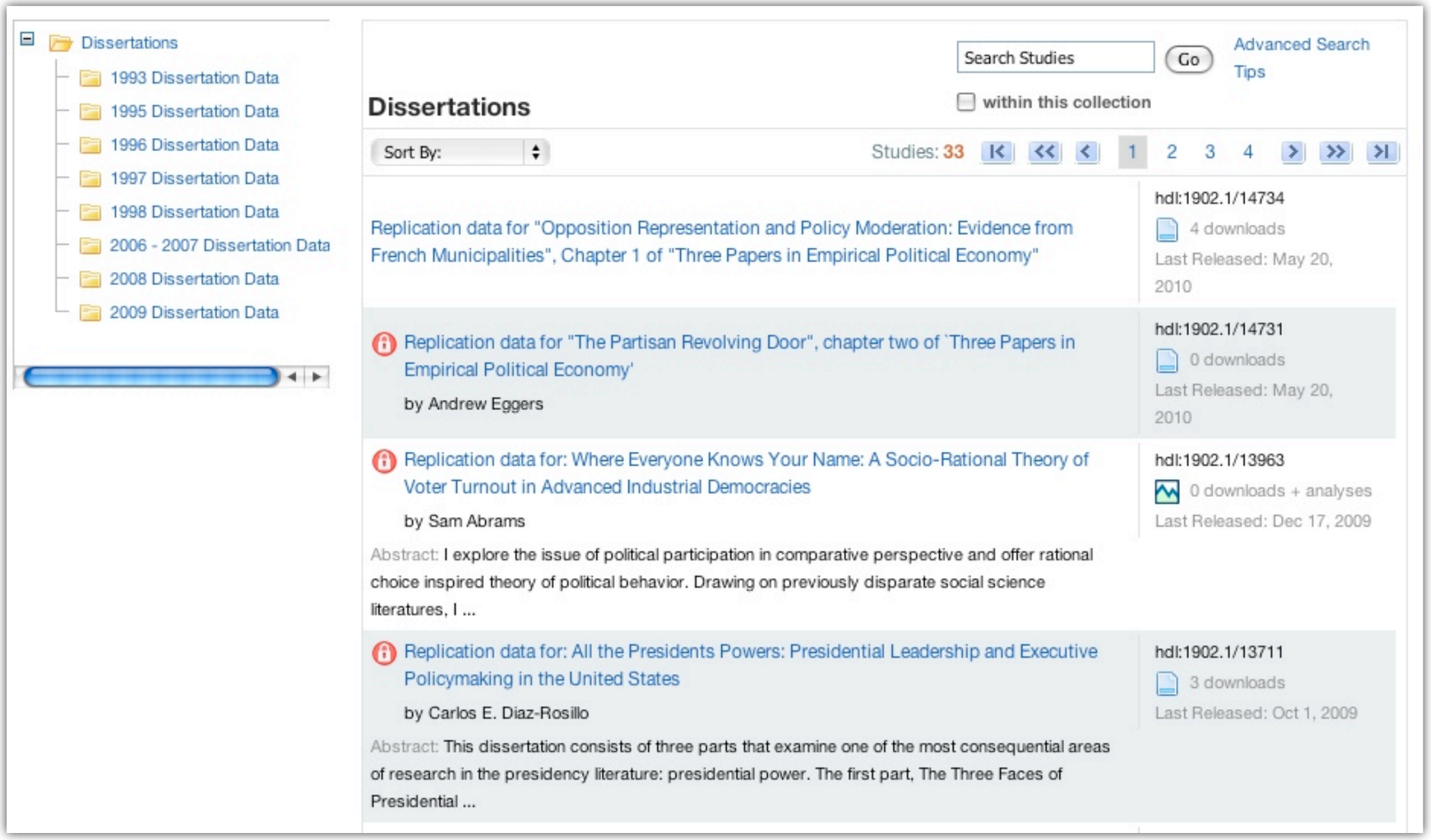
Select all files — Download All Selected Files

1. Documentation		
00092JamesBarnett.zip ZIP Archive - 562 KB - 18 downloads	Download	Codebooks and appendices for this study
00092James-Undergraduates-Codebooks.pdf Adobe PDF - 562 KB - 25 downloads	Download	Description of coded data variables
00092James-Undergraduates-Measures.pdf Adobe PDF - 1 MB - 2 downloads	Download	Collection of blank measures used in the study
2. Data		
00092James-Undergraduates-first_yr2.tab Tab Separated - 180 KB - 4 downloads + analyses TABULAR DATA | 480 Cases | 179 Variables	Restricted Access Access Subset/Analysis	Data file - first year students View Data Citation [+]
00092James-Undergraduates-seniors2.tab Tab Separated - 143 KB - 3 downloads + analyses TABULAR DATA | 325 Cases | 200 Variables	Restricted Access Access Subset/Analysis	Data file - seniors View Data Citation [+]
Text Files.zip ZIP Archive - 401 KB - 3 downloads	Restricted Access	Electronic text files of TATs

- Only authorized users or groups can access the files
- Groups can be based on IP address

☑ Distributes to the public, but also supports restricted access

... or the entire study can be restricted.



Restricted studies can be searched but user cannot access the study page unless has the right permissions

☑ Indexes all metadata for quick data discovery

Quick and advanced search of all metadata fields:

- Title
- Authors
- Distributors
- Abstract
- keywords
- Years
- Geographic information
- Data collection
- Terms of use
- ...
- Variable labels
- Variable descriptions

Uses Lucene index server as the search engine.

☑ Supports subsetting and analysis for (some) data files

REPLICATION DATA FOR: IMPROVING ANCHORING VIGNETTES: DESIGNING SURVEYS TO CORRECT INTERPERSONAL INCOMPARABILITY

Back to Study

DATA FILE: HOPKINSDATANOZIP.TAB

Download Subset | Recode & Case-Subset | Descriptive Statistics | Advanced Statistical Analysis

Selected Variables

ZIP
CASEID
SPEED1
SPEED2

For SPSS and STATA Files

- ✓ Choose a Statistical Model
- **Categorical Data Analysis**
  - Cross-Tabulation
- **Ecological Inference Models**
  - Hierarchical Multinomial-Dirichlet Ecological Inference Model for R x C Tables
- **Event Count Models**
  - Negative Binomial Reg for Event Count Dep Vars
  - Poisson Reg for Event Count Dep Vars
- **Models for Continous Bounded Dependent Variables**
  - Exponential Reg for Duration Dep Vars
  - Gamma Reg for Cont, Positive Dep Vars
  - Log-Normal Reg for Duration Dep Vars
  - Weibull Reg for Duration Dep Vars
- **Models for Continuous Dependent Variables**
  - Least Squares Reg for Cont Dep Vars
  - Linear regression for Left-Censored Dep Variable
- **Models for Dichotomous Dependent Variables**
  - Logistic Reg for Binary Dep Vars
  - Probit Reg for Binary Dep Vars
  - Rare Events Logistic Reg for Binary Dep Vars
- **Models for Ordinal Dependent Variables**
  - Ordinal Logistic Reg for Ordered Cat Dep Vars
  - Ordinal Probit Reg for Ordered Cat Dep Vars

Select variables from table below (sele

Show 10 Variables

Variable Information Table

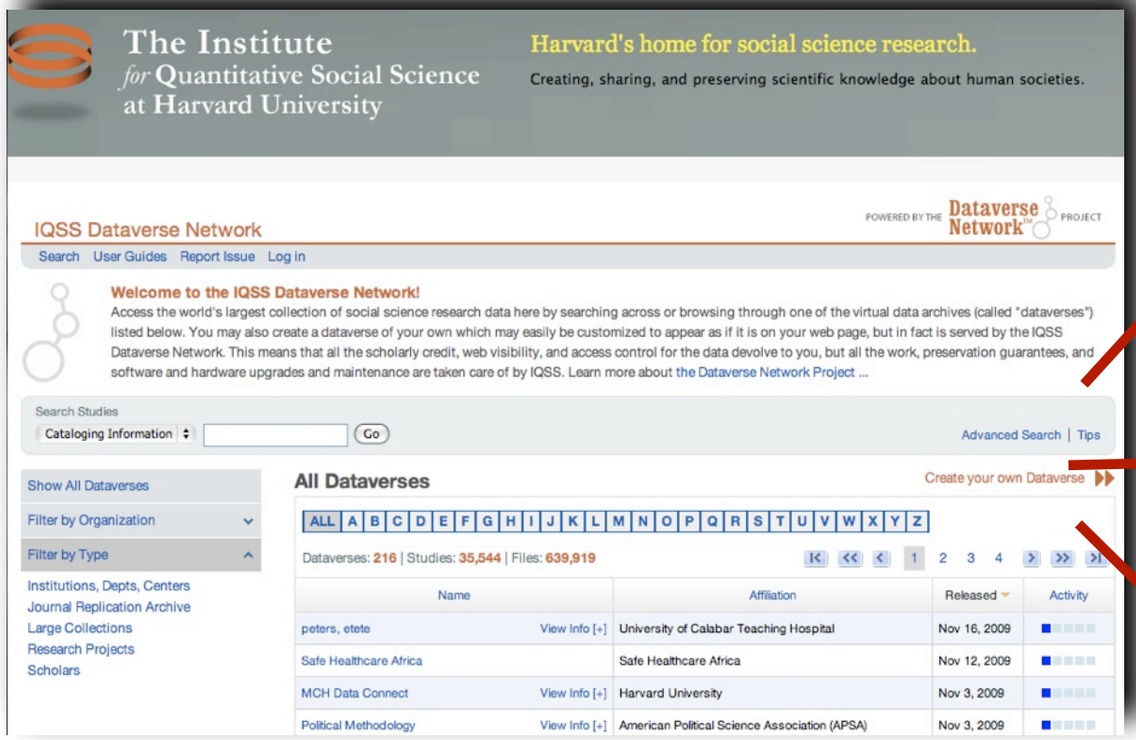
1 2 3 4

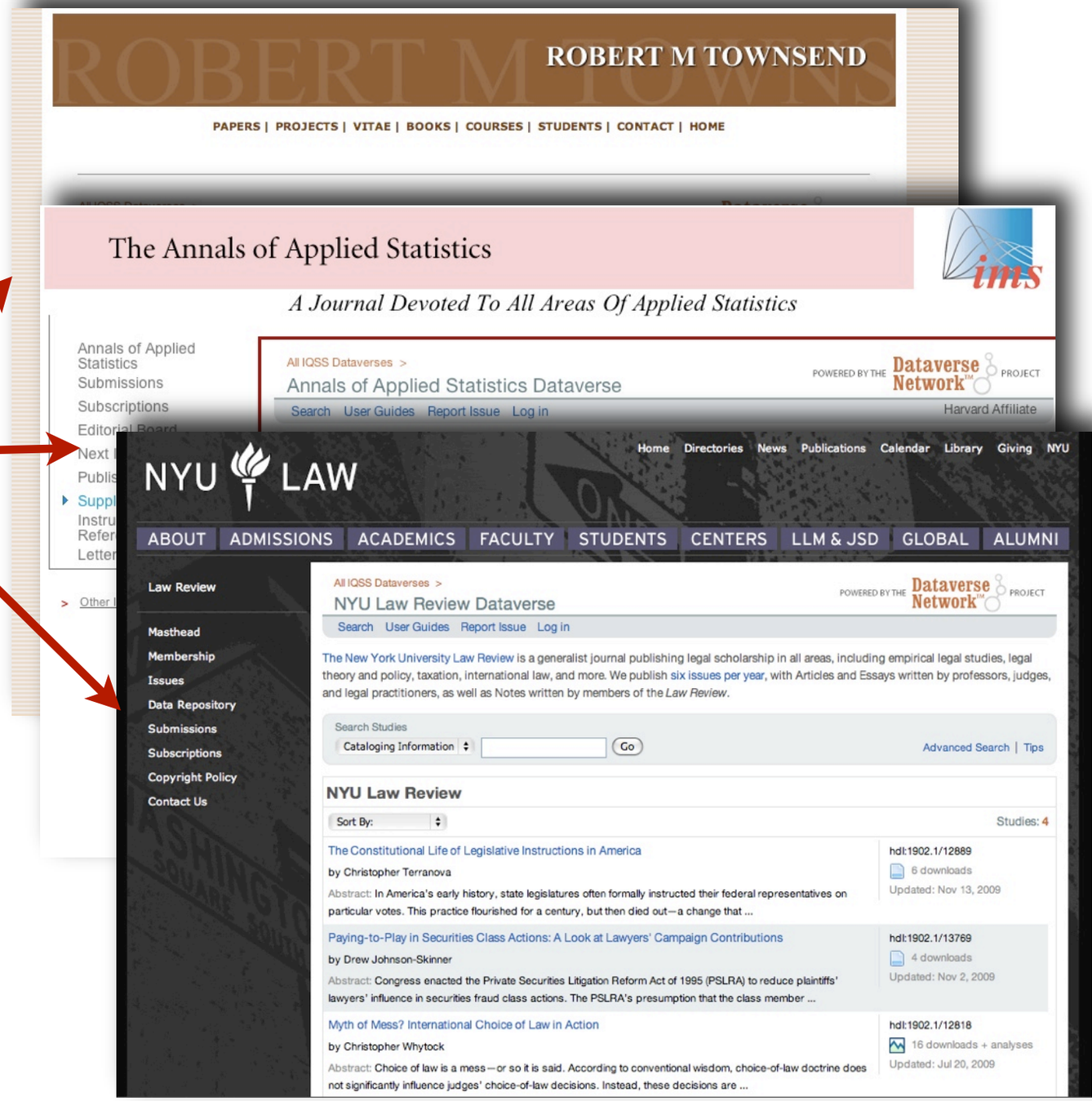
	Type			Summary
☑	Discrete			
☑	Discrete			
☐	Discrete	HOPRAND1	HOPRAND1	
☐	Discrete	MUCHSAY1	MUCHSAY1	

☑ Makes it easy to use and maintain

The Dataverse Network = the software

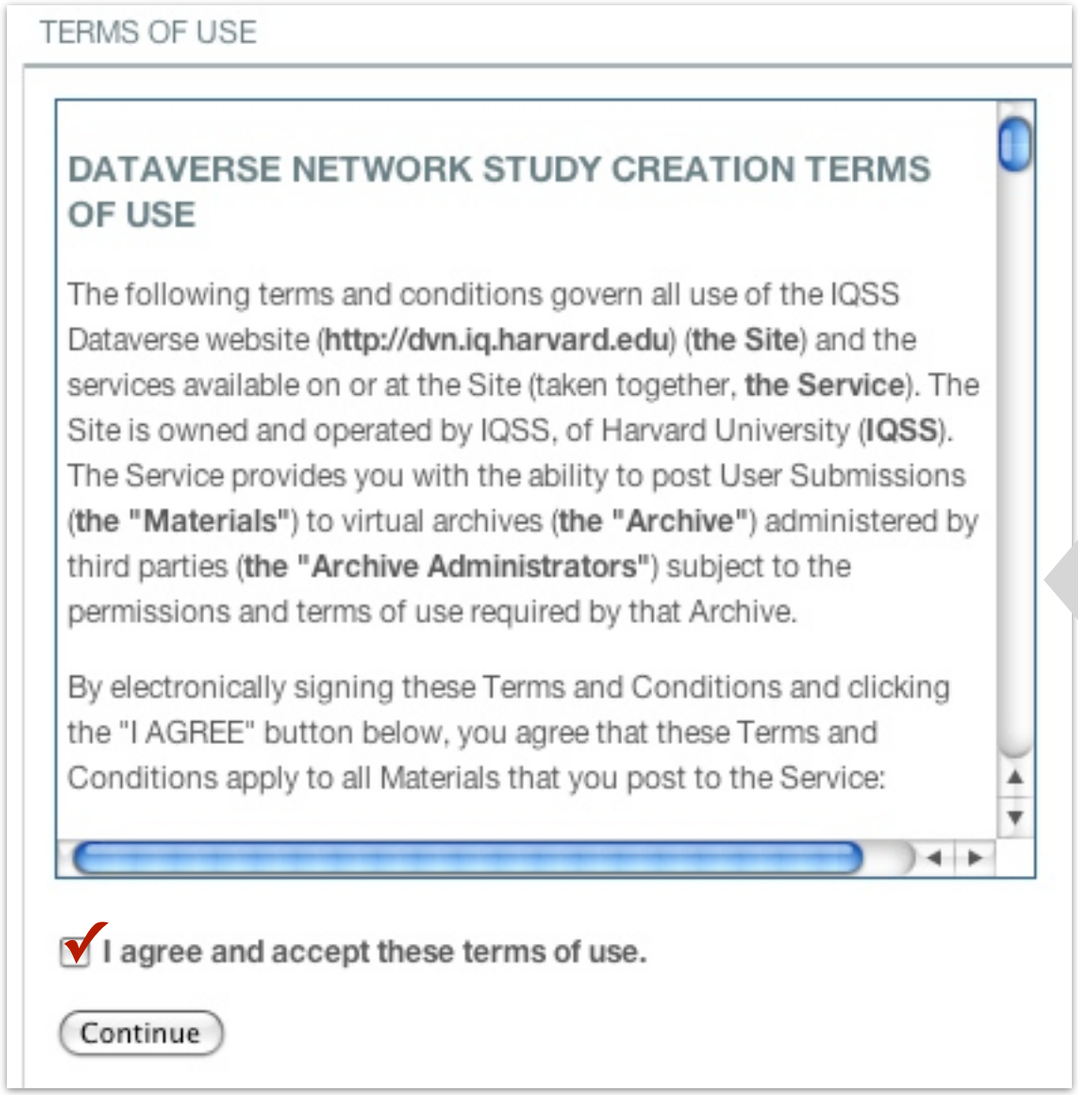
A dataverse = a site for publishing your data





- A Dataverse Network holds multiple dataverses
- Each dataverse can be created in two clicks and customized as you want
- Your dataverse may include your data or links to data from other dataverses

☑ Provides you Legal Protection



Agreement for Data Deposit

...

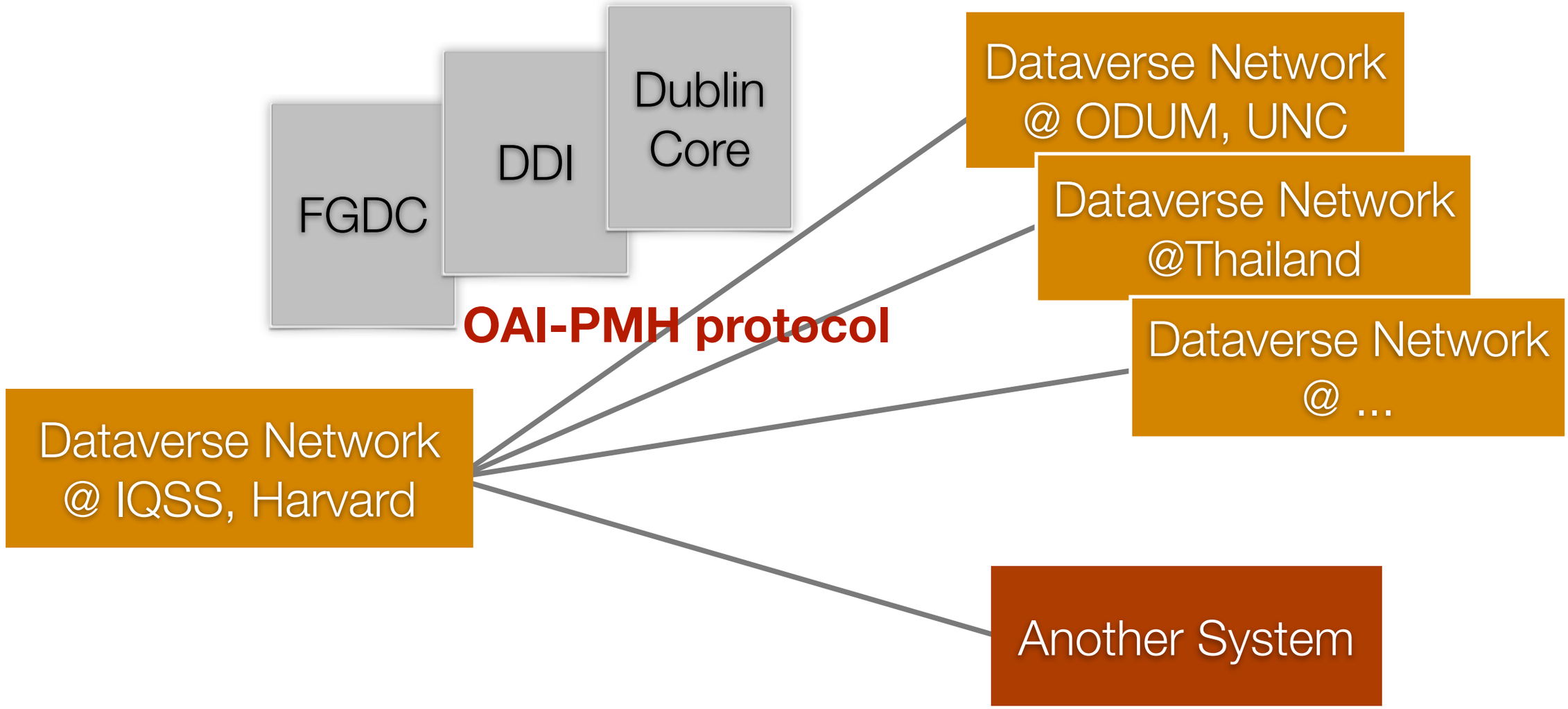
✓ If human subjects were studied in the collection of the Materials, you collected the Materials with IRB approval;

✓ The Materials do not contain high-risk confidential information such as social security numbers; credit card numbers; medical record numbers; health plan numbers; other account numbers; certificate/license numbers; or biometric identifiers (fingerprints, retina, voice print, DNA etc.).

...

- The terms of Use in the Harvard Dataverse Network are betted by lawyers from the Harvard's Office of the General Counsel
- You can customize the terms of use data deposit and data downloads in your dataverse or entire Dataverse Network

☑ Inter-operates with other systems



- Study and variable metadata is exported into XML (Dublin Core, Data Descriptive Initiative - DDI, and others) and MARC.
- Supports OAI-PMH, z39.50, LOCKKS
- Plan to add RDF support in the future

New Functionality

- Versioning: Never lose previous versions of studies
- Deaccessioning: Never lose track of previous holding
- Open access options: Enable contributors to collaboratively edit
- User comments: Collect user comments about data
- UNF 5.0: Advanced semantic fingerprints for your data
- Social Network Data: Deposit, format, analyze, and preserve social networks
- LOCKSS compatibility: Automatic replication for preservation

Thanks!

Next...

We invite you to participate in a workshop on Data Publication and Citation Principles at IQSS, Harvard University

Organizers: Micah Altman and Mercè Crosas

Contact me at mcrosas@hmdc.harvard.edu if you are interested.

http://thedata.org (The project site)

http://dvn.iq.harvard.edu (The Dataverse Network at Harvard)

Acknowledgements:

Gary King, Micah Altman, Gustavo Durand, Ellen Kraffmiller, Kevin Condon, Bob Treacy, Leonid Andrev, Michael Heppler, Steve Kraffmiller